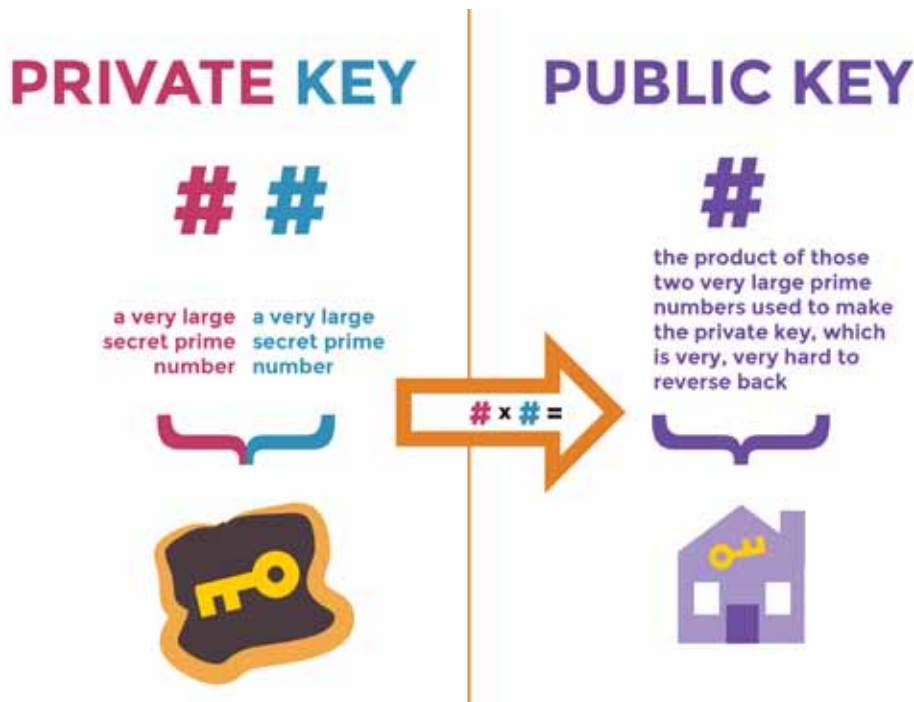


Public Key Infrastructure: An Introduction

Public key infrastructure helps achieve data confidentiality, integrity, and authentication, thus securing enterprises from cyber attacks. Let's get familiar with the many components of this infrastructure.



The simplest encryption method used in cryptography is a key, which is a random set of characters to mathematically convert your text to some junk content, which cannot be understood unless decrypted.

The next method used is a secret key, in which a single key is used for both encryption and decryption. Either party uses the same key, leading to faster performance. The challenge is to send the secret key before starting the encryption.

Public key and private key

In public key cryptography, we use a public and private key. There are two common use cases for this:

- To share confidential information, we encrypt the message using the public key of the receiver. This can be decrypted only by using the paired private key, which is with the receiver and never shared with anyone.
- To authenticate a message, one can encrypt it using its private key, which can be decrypted only by using the public key.

Public key infrastructure (PKI)

In public key cryptography, the concern is whether we are talking to the legitimate user, as anyone can have the pair of public and private keys. PKI consists of technologies and

policies that help build and use digital certificates, which can be validated by certificate authorities (CAs) using various models.

PKI protects information by providing:

- Identity authentication
- Integrity verification
- Privacy assurance
- Access authorisation
- Transaction authorisation
- Non-repudiation support

Certificate authorities (CAs)

The certificate authority or CA is an entity responsible for issuing and guaranteeing certificates. For secured communication, the CA must trust both